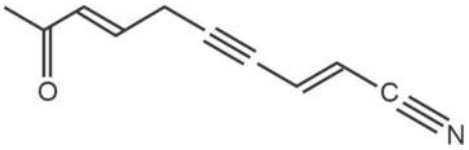
CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. In the given structure, number of sp and sp^2 hybridized carbon atoms present respectively are:

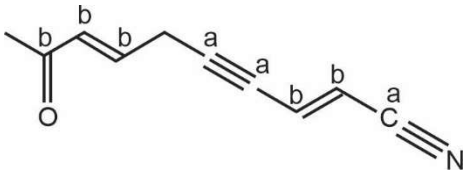


(1) 3 and 6 (2) 4 and 6

(3) 4 and 5 (4) 3 and 5

Answer (4)

Sol. The given structure is



Number of sp hybridized C-atoms(a) = 3

Number of sp^2 hybridized C-atoms(b) = 5

52. Identify correct statement/s:

(A) $-OCH_3$ and $-NHCOCH_3$ are activating group.

(B) –CN and –OH are meta directing group.

(C) –CN and $-SO_3H$ are meta directing group.

(D) Activating groups act as ortho- and para directing groups.

(E) Halides are activating groups.

Choose the correct answer from the options given below:

(1) (A), (C) and (D) only

(2) (A) and (C) only

(3) (A) only

(4) (A), (B) and (E) only

Answer (1)

Sol. $-OCH_3$, $NHCOCH_3$ and –OH are activating groups because the atom directly bonded to benzene ring activates the ring by +R effect using its lone pair of electrons. Activating groups are ortho- and para directing groups.

–CN and $-SO_3H$ are deactivating groups because the atom directly bonded to benzene is bonded to more electronegative atom through multiple bonds and they are meta directing groups due to –R effect. Halides are deactivating groups due to –I effect.

Correct statements are (A), (C) and (D) only.

53. The conditions and consequence that favours the $t_{2g}^3 e_g^1$ configuration in a metal complex are:

(1) strong field ligand, high spin complex

(2) weak field ligand, high spin complex

(3) weak field ligand, low spin complex

(4) strong field ligand, low spin complex

Answer (2)

Sol. The conditions and consequence that favour $t_{2g}^3 e_g^1$ configuration in a metal complex are

(i) weak field ligand, and

(ii) high spin complex

For weak field ligands, splitting energy (Δ_0) is lower than pairing energy (P). As a result distribution of electrons for $3d^4$ will be $t_{2g}^3 e_g^1$. It results in high spin complex due to maximum number of unpaired electrons.



54. Given below are two statements:

Statement (I): The first ionization energy of Pb is greater than that of Sn.

Statement (II): The first ionization energy of Ge is greater than that of Si.

In the light of the above statements, choose the correct answer from the options given below:

(1) **Statement I** is false but **Statement II** is true
(2) **Statement I** is true but **Statement II** is false
(3) Both **Statement I** and **Statement II** are false
(4) Both **Statement I** and **Statement II** are true

Answer (2)

Sol. First ionization energy of Lead = 715 kJ mol^{-1}

First ionization energy of Tin = 708 kJ mol^{-1}

(IE_1) of Lead is greater than that of Tin due to ineffective shielding of d- and f-electrons.

Therefore **Statement-I** is true.

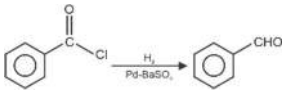
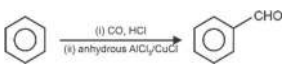
First ionization energy of Germanium = 761 kJ mol^{-1}

First ionization energy of Silicon = 786 kJ mol^{-1}

(IE_1) of Germanium is lower than that of Silicon as the effect of higher atomic radius of Ge outweighs the increase in nuclear charge from Si to Ge and effective shielding of inner electrons.

Therefore **Statement-II** is false.

55. Match **List-I** with **List-II**.

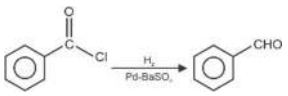
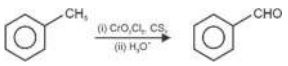
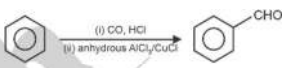
	List-I		**List-II**
(A)	$RCN \xrightarrow[(ii)\ H_3O^+]{(i)\ SnCl_2,\ HCl} RCHO$	(I)	Etard reaction
(B)		(II)	Gatterman-Koch reaction
(C)		(III)	Rosenmund reduction
(D)		(IV)	Stephen reaction

Choose the **correct** answer from the options given below:

(1) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
(2) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(3) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
(4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Answer (3)

Sol.

	List-I		**List-II**
(A)	$RCN \xrightarrow[(ii)\ H_3O^+]{(i)\ SnCl_2,\ HCl} RCHO$	(IV)	Stephen reaction
(B)		(III)	Rosenmund reduction
(C)		(I)	Etard reaction
(D)		(II)	Gatterman-Koch reaction

Correct match: (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

56. The successive 5 ionisation energies of an element are 800, 2427, 3658, 25024 and 32824 kJ/mol, respectively. By using the above values predict the group in which the above element is present:

(1) Group 14 (2) Group 4
(3) Group 13 (4) Group 2

Answer (3)

Sol. The successive 5 ionisation energies of an element are 800, 2427, 3658, 25024 and 32824 kJ/mol, respectively.

$$\frac{IE_2}{IE_1} = \frac{2427}{800} = 3.03$$

$$\frac{IE_3}{IE_2} = \frac{3658}{2427} = 1.51$$

$$\frac{IE_4}{IE_3} = \frac{25024}{3658} = 6.84$$

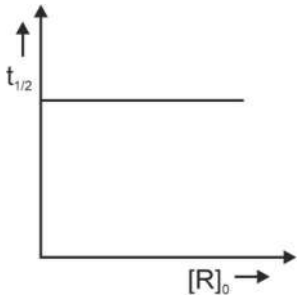


$$\frac{IE_5}{IE_4} = \frac{32824}{25024} = 1.31$$

Since (IE_4 / IE_3) value is maximum, the element belongs to group 13.

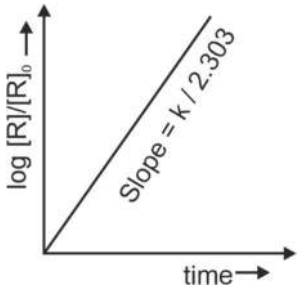
57. Given below are two statements:

Statement (I):



is valid for first order reaction.

Statement (II):



is valid for first order reaction.

In the light of the above statements, choose the correct answer from the options given below:

(1) **Statement I** is true but **Statement II** is false

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is false but **Statement II** is true

(4) Both **Statement I** and **Statement II** are true

Answer (1)

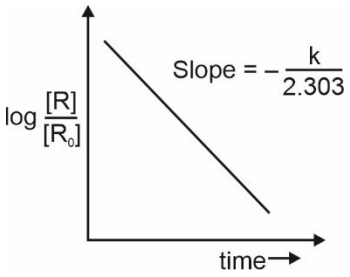
Sol. Half life ($t_{1/2}$) of a first order reaction does not depend on initial concentration of the reactant [R_0]. Therefore **Statement I** is true. Rate constant (k) of a first order reaction is given by

$$k = \frac{2.303}{t}\log\frac{[R_0]}{[R]}$$

$$\log\frac{[R]}{[R_0]} = -\left(\frac{k}{2.303}\right)t$$

Plot of $\log\frac{[R]}{[R_0]}$ versus time is linear having

$$\text{slope} = -\frac{k}{2.303}$$



$\therefore$ **Statement II** is false.

58. Given below are two statements:

Statement (I): Experimentally determined oxygen-oxygen bond lengths in the O_3 are found to be same and the bond length is greater than that of a O = O (double bond) but less than that of a single (O – O) bond.

Statement (II): The strong lone pair-lone pair repulsion between oxygen atoms is solely responsible for the fact that the bond length in ozone is smaller than that of a double bond (O = O) but more than that of a single bond (O – O).

In the light of the above statements, choose the correct answer from the options given below :

(1) **Statement I** is false but **Statement II** is true

(2) Both **Statement I** and **Statement II** are true

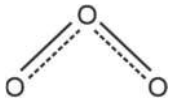
(3) **Statement I** is true but **Statement II** is false

(4) Both **Statement I** and **Statement II** are false



Answer (3)

Sol.

Bond	(O – O) Bond order
(O – O)	1.0
	1.5
O = O	2.0

$$\text{Bond length} \propto \frac{1}{\text{Bond order}}$$

Order of O – O bond length

$O = O < O_3 < O - O$

$\therefore$ **Statement I** is true.

Lone pair-lone pair repulsion between O-atoms is not solely responsible for the correct order of O–O bond length. Bond order also should be considered.

$\therefore$ **Statement II** is false.

59. When Ethane-1,2-diamine is added progressively to an aqueous solution of Nickel (II) chloride, the sequence of colour change observed will be:

(1) Green $\rightarrow$ Pale blue $\rightarrow$ Blue $\rightarrow$ Violet

(2) Pale blue $\rightarrow$ Blue $\rightarrow$ Violet $\rightarrow$ Green

(3) Pale blue $\rightarrow$ Blue $\rightarrow$ Green $\rightarrow$ Violet

(4) Violet $\rightarrow$ Blue $\rightarrow$ Pale blue $\rightarrow$ Green

Answer (1)

Sol. If ethane-1,2-diamine (en) is added progressively to an aqueous solution of $NiCl_2$ in the molar ratios en : Ni 1 : 1, 2 : 1 and 3 : 1, the following series of reactions and their associated colour changes occur. [Ref : NCERT (XII) Part-1, p 260]

$$\underset{\text{Green}}{[Ni(H_2O)_6]^{2+}(aq)} + en(aq) \longrightarrow$$

$$\underset{\text{Pale blue}}{[Ni(H_2O)_4\,en]^{2+}(aq)} + 2H_2O$$

$$[Ni(H_2O)_4\,en]^{2+}(aq) + en(aq) \longrightarrow$$

$$\underset{\text{Blue/Purple}}{[Ni(H_2O)_2(en)_2]^{2+}(aq)} + 2H_2O$$

$$[Ni(H_2O)_2(en)_2]^{2+}(aq) + en(aq) \longrightarrow$$

$$\underset{\text{Violet}}{[Ni(en)_3]^{2+}(aq)} + 2H_2O$$

60. Which of the following mixing of 1 M base and 1 M acid leads to the largest increase in temperature?

(1) 30 mL HCl and 30 mL NaOH

(2) 30 mL CH_3COOH and 30 mL NaOH

(3) 45 mL CH_3COOH and 25 mL NaOH

(4) 50 mL HCl and 20 mL NaOH

Answer (1)

Sol. The rise in temperature of neutralization reaction will be maximum for maximum number of moles of strong acid and strong base neutralized and lower volume of final solution.

(1) $HCl + NaOH \rightarrow NaCl + H_2O$

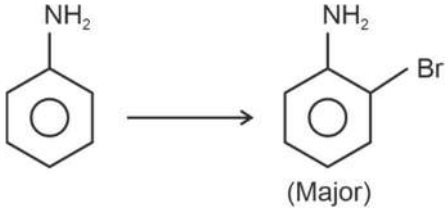
mmol 30 30

- -

Final volume of solution = 60 mL

Option (2) and (3) have weak acids and in option (4) only 20 mmol of HCl will be neutralized with 70 mL final volume.

61. For reaction



(Major)

The correct order of set of reagents for the above conversion is :

(1) Br_2 | $FeBr_3$, $H_2O(\Delta)$, NaOH

(2) Ac_2O, Br_2, $H_2O(\Delta)$, NaOH

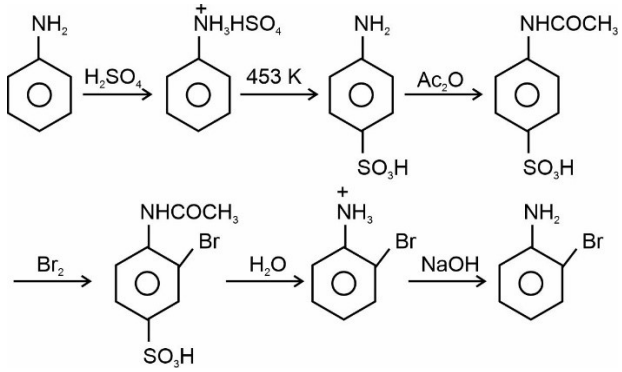
(3) H_2SO_4, Ac_2O, Br_2, $H_2O(\Delta)$, NaOH

(4) Ac_2O, H_2SO_4, Br_2, NaOH

Answer (3)



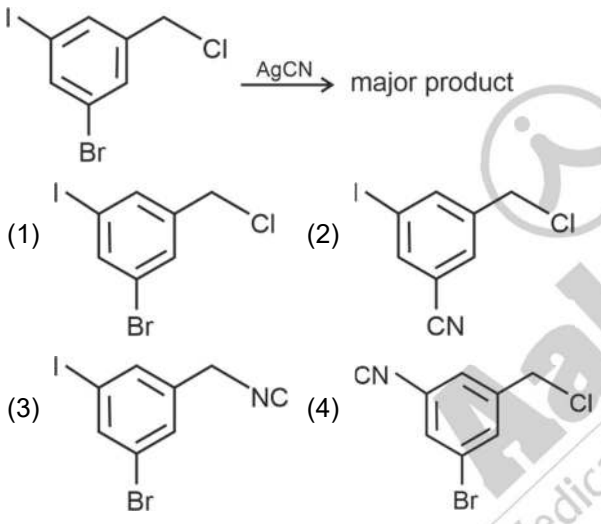
Sol.



Correct sequence of reagents

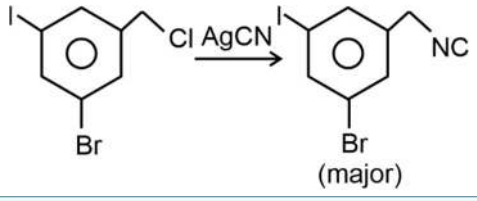
H_2SO_4, Ac_2O, Br_2, $H_2O(\Delta)$, NaOH

62. The structure of the major product formed in the following reaction is :



Answer (3)

Sol. Haloalkanes react with AgCN to give isocyanide as major product and haloarenes do not react with AgCN.



63. Find the compound 'A' from the following reaction sequences.

$$A \xrightarrow{\text{aqua-regia}} B \xrightarrow[(2)\ AcOH]{(1)\ KNO_2[NH_4OH]} \text{yellow ppt}$$

(1) CoS (2) MnS

(3) NiS (4) ZnS

Answer (1)

Sol. Compound (A) in the given reaction sequence is likely to be CoS.

$$CoS + HNO_3 + 3HCl \rightarrow Co^{2+} + S\downarrow + NOCl\uparrow + 2Cl^- + 2H_2O$$

The above solution is neutralised with NH_4OH.

To a neutral solution of Co^{2+}, acetic acid and saturated solution of KNO_2 are added which results in the formation of yellow precipitate of $K_3[Co(NO_2)_6]$

$$Co^{2+} + 7NO_2^- + 2H^+ + 3K^+ \rightarrow K_3[Co(NO_2)_6] + NO\uparrow + H_2O$$

64. The elemental composition of a compound is 54.2% C, 9.2% H and 36.6% O. If the molar mass of the compound is 132 g mol^{-1}, the molecular formula of the compound is:

[Given: The relative atomic mass of C : H : O = 12 : 1 : 16]

(1) $C_6H_{12}O_6$ (2) $C_4H_8O_2$

(3) $C_4H_9O_3$ (4) $C_6H_{12}O_3$

Answer (4)

Sol.

Element	Mass %	Mole %	Molar ratio
C	54.2	4.52	2
H	9.2	9.2	4
O	36.6	2.28	1

Empirical formula of compound is C_2H_4O



Molecular mass of compound = 132 g mol^{-1}

Molecular formula of compound is $(C_2H_4O)_n$

$$n = \frac{\text{Molecular mass}}{\text{EF mass}} = \frac{132}{44} = 3$$

$\therefore$ Molecular formula of compound is $C_6H_{12}O_3$

65. For hydrogen atom, the orbital/s with lowest energy is/are:

(A) 4s

(B) $3p_x$

(C) $3d_{x^2-y^2}$

(D) $3d_{z^2}$

(E) $4p_z$

Choose the **correct** answer from the options given below:

(1) (A) and (E) only

(2) (A) only

(3) (B) only

(4) (B), (C) and (D) only

Answer (4)

Sol. For hydrogen atom and one electron species, the energy of orbitals is decided by the value of principal quantum number. Higher the value of principal quantum number, higher will be the energy of orbital.

(A) 4s n = 4

(B) $3p_x$ n = 3

(C) $3d_{x^2-y^2}$ n = 3

(D) $3d_{z^2}$ n = 3

(E) $4p_z$ n = 4

$\therefore$ (B), (C) and (D) have orbitals with the lowest energy.

66. Based on the data given below:

$E^\circ_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33V$ $E^\circ_{Cl_2/Cl^-} = 1.36V$

$E^\circ_{MnO_4^-/Mn^{2+}} = 1.51V$ $E^\circ_{Cr^{3+}/Cr} = -0.74V$

The strongest reducing agent is:

(1) Cl^- (2) Cr

(3) MnO_4^- (4) Mn^{2+}

Answer (2)

Sol. $E^\circ_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33V$ $E^\circ_{Cl_2/Cl^-} = 1.36V$

$E^\circ_{MnO_4^-/Mn^{2+}} = 1.51V$ $E^\circ_{Cr^{3+}/Cr} = -0.74V$

The species which has the most negative value of standard reduction potential will be the strongest reducing agent. Since Cr^{3+}/Cr has SRP value of –0.74V, Cr is the strongest reducing agent.

67. Match List – I with List – II

List – I (Transition metal ion)		List – II (Spin only magnetic moment (B.M.))	
(A)	Ti^{3+}	(I)	3.87
(B)	V^{2+}	(II)	0.00
(C)	Ni^{2+}	(III)	1.73
(D)	Sc^{3+}	(IV)	2.84

Choose the correct answer from the options given below:

(1) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

(2) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(3) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

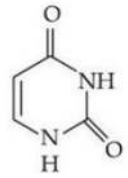
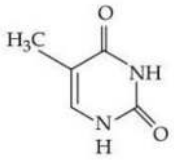
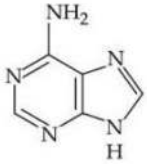
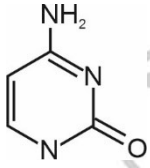
Answer (3)

Sol.

	Transition Metal ion	Electronic configuration	No. of unpaired electrons	Magnetic moment (BM)
(A)	Ti^{3+}	$3d^1$	1	1.73
(B)	V^{2+}	$3d^3$	3	3.87
(C)	Ni^{2+}	$3d^8$	2	2.84
(D)	Sc^{3+}	$3d^o$	0	0.00



68. Match List-I with List-II

List-I		List-II	
(A)	Adenine	(I)	
(B)	Cytosine	(II)	
(C)	Thymine	(III)	
(D)	Uracil	(IV)	

Choose the correct answer from the options given below:

(1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

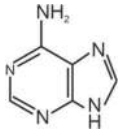
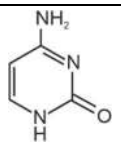
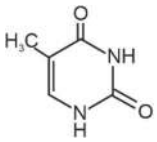
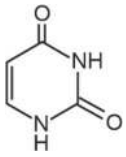
(2) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(3) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Answer (3)

Sol.

Compound		Structure	
(A)	Adenine	III.	
(B)	Cytosine	IV.	
(C)	Thymine	II.	
(D)	Uracil	I.	

69. $S(g)+\frac{3}{2}O_2(g) \rightarrow SO_3(g)+2x$ kcal

$SO_2(g)+\frac{1}{2}O_2(g) \rightarrow SO_3(g)+y$ kcal

The heat of formation of $SO_2(g)$ is given by:

(1) $\frac{2x}{y}$ kcal

(2) $y - 2x$ kcal

(3) $x + y$ kcal

(4) $2x + y$ kcal

Answer (2)

Sol.

(i) $S(g)+\frac{3}{2}O_2(g) \rightarrow SO_3(g)+2x$ kcal ΔH_1

(ii) $SO_2(g)+\frac{1}{2}O_2(g) \rightarrow SO_3(g)+y$ kcal ΔH_2

(i) – (ii)

$S(g)+O_2(g) \rightarrow SO_2(g)$ ΔH

$\Delta H = \Delta H_1 - \Delta H_2$

$= -2x - (-y) = (y - 2x)$ kcal

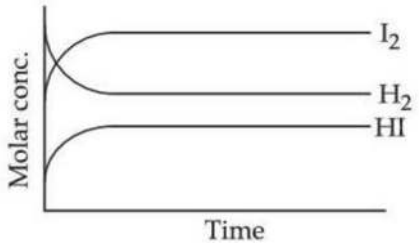


70. For the reaction,

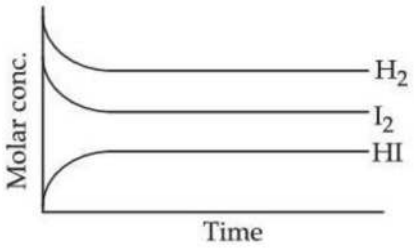
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

Attainment of equilibrium is predicted correctly by:

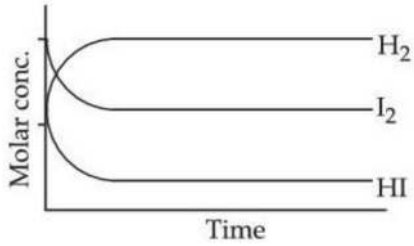
(1)



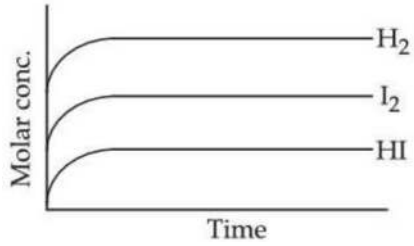
(2)



(3)



(4)

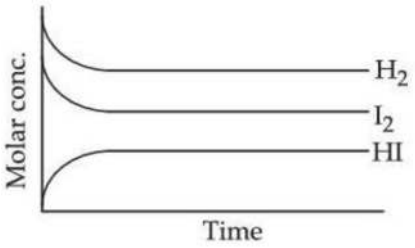


Answer (2)

Sol.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

Concentrations of $H_2(g)$ and $I_2(g)$ decrease with time while concentration of HI(g) increases with time. At equilibrium $H_2(g)$, $I_2(g)$ and HI(g) attain constant values. Correct plot of molar concentration with time is



71. Consider a complex reaction taking place in three steps with rate constants k_1, k_2 and k_3 respectively. The overall rate constant k is given by the expression $k = \sqrt{\frac{k_1 k_3}{k_2}}$. If the activation energies of the three steps are 60, 30 and 10 kJ mol^{-1} respectively, then the overall energy of activation in kJ mol^{-1} is _____. (Nearest integer)

Answer (20)

Sol. k_1, k_2 and k_3 are given as rate constants of three steps of a complex reaction. Rate constant (k) of the overall reaction is given as

$$k = \sqrt{\frac{k_1 k_3}{k_2}}$$

Activation energies of the three steps are given as

E_{a_1} = 60 kJ mol^{-1}, E_{a_2} = 30 kJ mol^{-1},

E_{a_3} = 10 kJ mol^{-1}

From Arrhenius equation, we know that

$k = Ae^{-Ea/RT}$

If E_a is the activation energy of the overall reaction, then

$$E_a = \frac{1}{2}\left[E_{a_1} + E_{a_3} - E_{a_2}\right]$$

$$= \frac{1}{2}[60 + 10 - 30] = 20 \text{ kJ mol}^{-1}$$



72. In Carius method of estimation of halogen, 0.25 g of an organic compound gave 0.15 g of silver bromide (AgBr). The percentage of Bromine in the organic compound is _____ × 10^{-1} % (Nearest integer).

(Given: Molar mass of Ag is 108 and Br is 80 g mol^{-1})

Answer (255)

Sol. Mass of organic compound = 0.25 g

Mass of AgBr = 0.15 g

No. of moles of Br = No. of moles of AgBr = $\frac{0.15}{188}$

Mass of Br = $\frac{0.15 \times 80}{188}$ g

% of Br = $\frac{0.15 \times 80 \times 100}{188 \times 0.25}$

= 25.5 %

= 255 × 10^{-1} %

73. The observed and normal molar masses of compound MX_2 are 65.6 and 164 respectively. The percent degree of ionisation of MX_2 is _____ %. (Nearest integer)

Answer (75)

Sol. Normal molar mass of MX_2 = 164.0 g mol^{-1}

Observed molar mass of MX_2 = 65.6 g mol^{-1}

Van't Hoff factor (i) = $\frac{\text{Normal molar mass}}{\text{Observed molar mass}}$

$= \frac{164}{65.6} = 2.5$

If α is the degree of ionisation, then

$$\underset{1-\alpha}{MX_2} \rightleftharpoons \underset{\alpha}{M^{2+}} + \underset{2\alpha}{2X^-}$$

$i = 1 - \alpha + \alpha + 2\alpha = 1 + 2\alpha$

$1 + 2\alpha = 2.5$

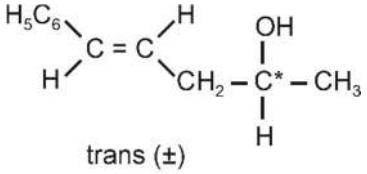
$\alpha = 0.75$

$\therefore$ Percent degree of ionisation = 75%

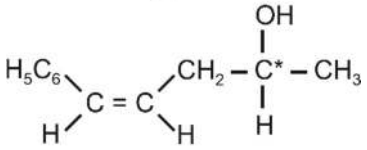
74. The possible number of stereoisomers for 5-phenylpent-4-en-2-ol is _____.

Answer (4)

Sol. The given compound is 5-phenylpent-4-en-2-ol. Possible stereoisomers are



trans (±)



cis (±)

$\therefore$ 4 possible stereoisomers

75. The hydrocarbon (X) with molar mass 80 g mol^{-1} and 90% carbon has _____ degree of unsaturation.

Answer (3)

Sol.

Element	Mass %	Mole %	Molar ratio
C	90	7.5	1
H	10	10	1.33

$\therefore$ Empirical formula is C_3H_4

Molecular mass = 80 g mol^{-1}.

Molecular formula = $(C_3H_4)_n$

$n = \frac{\text{Molecular mass}}{\text{EF mass}} = \frac{80}{40} = 2$

$\therefore$ Molecular formula is C_6H_8

Degree of unsaturation = $\frac{2 \times 6 + 2 - 8}{2} = 3$

❑ ❑ ❑

